



AI Risk Taxonomy for Financial Services

Complete risk register template and assessment methodology

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Contact: advisor@rodney-ai.com

Table of Contents

1. Executive Summary
2. Risk Categories
3. Risk Assessment Methodology
4. AI Risk Register Template
5. Risk Indicators & Triggers
6. Mitigation Strategies
7. Regulatory Alignment

1. Executive Summary

AI systems in financial services introduce new and amplified risks across model, data, operational, and regulatory dimensions. This taxonomy provides a comprehensive framework for identifying, assessing, and managing these risks in a consistent manner across the enterprise.

2. Risk Categories

Model Risk

Model Performance Risk: Model accuracy degrades or fails to perform as expected in production **Model Bias Risk:** Systematic favoritism toward certain groups or outcomes **Model Interpretability Risk:** Model decisions cannot be explained or justified **Model Stability Risk:** Model behaves unpredictably with small input changes

Data Risk

Data Quality Risk: Training or production data is incomplete, inaccurate, or unrepresentative **Data Bias Risk:** Historical biases in data perpetuate or amplify systemic biases **Data Privacy Risk:** Personally identifiable information is exposed or misused **Data Governance Risk:** Unclear data ownership or inconsistent definitions

Operational Risk

Integration Risk: AI system fails to integrate properly with existing systems **Governance Risk:** Insufficient controls and oversight of AI systems **Vendor Risk:** Third-party AI providers lack adequate controls **Change Management Risk:** Organization unable to adopt and use AI systems effectively

Regulatory & Compliance Risk

Fair Lending Risk: AI systems discriminate based on protected characteristics **Consumer Protection Risk:** AI system fails to protect consumer interests **Explainability Risk:** Regulators cannot validate AI system decisions **Regulatory Change Risk:** New regulations make existing systems non-compliant

3. Risk Assessment Methodology

Risk Scoring Matrix

Risk Level = Likelihood × Impact

Rating	Likelihood	Impact to Business
1 - Low	<10% probability in 12 months	<\$100K loss / <1% operational impact
2 - Medium	10-50% probability in 12 months	\$100K-\$1M loss / 1-5% operational impact
3 - High	50-90% probability in 12 months	\$1M-\$10M loss / 5-20% operational impact
4 - Critical	>90% probability in 12 months	>\$10M loss / >20% operational impact

4. AI Risk Register Template

Each AI system should have a documented risk register with minimum the following entries:

Risk ID	Risk Category	Description	Current State	Likelihood	Impact	Overall Risk	Owner	Mitigation
AIR-001	Model Performance	Accuracy decline in validation set	Green	Low	Medium	Medium	Data Lead	Weekly monitoring
AIR-002	Data Bias	Historical bias in training data	Amber	Medium	High	High	Analytics Lead	Synthetic data augmentation
AIR-003	Regulatory	New fair lending requirements	Amber	High	Critical	Critical	Compliance	Impact assessment

5. Risk Indicators & Triggers

Model Performance Indicators

• Accuracy drops >5% from baseline • Precision or recall declines >10% • F1 score below minimum threshold • Error rates exceed SLA • Unexpected patterns in residuals

Data Quality Indicators

• Missing value rates >20% • Outlier prevalence increases >2x • Feature distributions shift >10% • Data freshness violations • Unexplained data changes

6. Mitigation Strategies

Model Risk Mitigation

- Regular model validation and backtesting
- Bias testing on protected attributes
- Model explainability assessments
- Gradual rollout with monitoring
- Regular retraining schedules

Data Risk Mitigation

- Data quality frameworks and SLAs
- Data audits for bias and representation
- Privacy-preserving techniques (anonymization, differential privacy)
- Data governance policies and metadata

■ Regular data lineage documentation

7. Regulatory Alignment

This taxonomy aligns with regulatory guidance from: • Federal Reserve SR 11-7 (Model Risk Management) • OCC Bulletin 2020-4 (AI/ML Governance) • FDIC Guidelines on Model Risk Management • Interagency Guidance on Fairness in Credit

Next Steps

This guide is designed to be a practical, hands-on resource for implementation. Use the templates and checklists as starting points for your organization's specific needs.

We're here to help.

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Email: advisor@rodney-ai.com

For consultation on implementation, governance, and strategic deployment of AI initiatives.

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